

Unit 12, Lesson 6: Problem Solving with Volume of Cylinders

Benchmark

MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders.

Learning Target

- ☐ I can solve real-world problems involving the volume of right circular cylinders.

12.6.1 Warm-Up: Notice or Wonder

1. Two cylindrical cans are shown.

A close-up picture of each can shows that both cans contain 15 ounces of food.

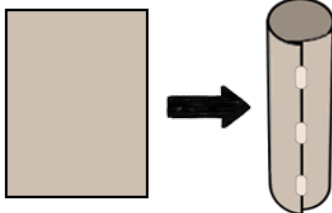
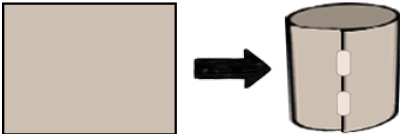


- a. Explain how it could be possible for two differently shaped cans to have the same volume.
- b. Describe how the dimensions of the two cans compare to each other.

Dimension	How do they compare?
radius	
height	

12.6.2 Exploration: Popcorn Containers

1. Dan is using paper to create a popcorn container. He can make two differently shaped cylindrical containers by rolling two identical pieces of paper. Container A is taller and narrower, while Container B is shorter and wider.

Container A	Container B
	

- a. Which container do you think will hold more popcorn?
- b. Without having popcorn to test, what information would be helpful in determining which container holds more popcorn?

The dimensions of each piece of paper prior to rolling it to form a cylinder are shown.

Container A	Container B
	

2. The radius of each resulting cylinder is given in the table. Complete the statements.

Container A	Container B
- The height of the paper is the ○ height ○ circumference of the cylinder. - The width of the paper is the ○ height ○ circumference of the cylinder. - The radius of the cylinder is 1.354 inches (in.).	- The height of the paper is the ○ height ○ circumference of the cylinder. - The width of the paper is the ○ height ○ circumference of the cylinder. - The radius of the cylinder is 1.752 in.

3. Calculate the volume of each container. Use 3.14 for π and round to the nearest thousandth.

Volume of Container A	Volume of Container B

4. Complete the statement.

Container

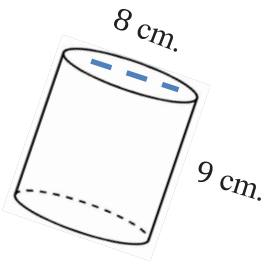
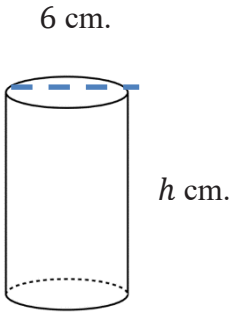
○ A

○ B

 can hold more popcorn.

12.6.3 Guided Instruction: Using the Volume Formula to Problem-Solve

1. Cylinder C and Cylinder D have the same volume in cubic centimeters (cm³).

Cylinder C	Cylinder D
	

a. Find the volume of Cylinder C. Leave it in terms of π .

b. Write an expression for the volume of Cylinder D.

$V =$ _____

c. Write an equation to show the volumes are equal to each other.

_____ = _____

d. Solve the equation.

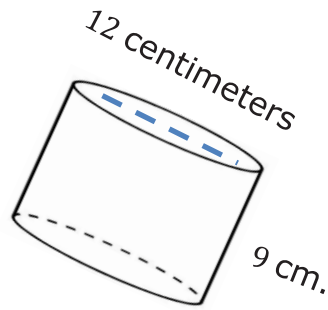
e. What does the solution to the equation represent?

12.6.4 Try It: Finding the Height

1. Julius wants to build an aquarium for his pet goldfish in the shape of a right circular cylinder. He read that his fish needs to live in 1538.6 cubic inches (in.³) of water. If his aquarium has a diameter of 14 in., what is the height of the aquarium? Use 3.14 for π .

12.6.5 Your Turn!

1. Find the volume of the cylinder shown. Write your answer in terms of π .



2. The ceramic cylinder shown is advertised as holding 71.5 cubic inches. How tall must the ceramic cylinder be? Use 3.14 for π and round your answer to the nearest thousandth.

$$d = 4.5 \text{ in.}$$



3. Ruby is building a new above-ground pool for her backyard. The manual says the volume of the pool is 21.613 cubic meters (m^3) and has a diameter of 4.6 m. What is the height of the new pool? Use $\frac{22}{7}$ for π and round to the nearest tenth.



12.6.6 Wrap-Up: Reflection

1. Place an **X** on the line to indicate your understanding of each learning target. Then, provide a brief explanation or evidence to support your self-assessment.

Learning Target	Self-Assessment		
I can find the volume of a cylinder.	I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
Evidence			
I can apply the volume formula to solve real-world problems.	I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
Evidence			